Bioinformatics Project - Predicting circRNA-miRNA interaction using RNA-FM and FFNN

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1 Introduction

The problem. Ribonucleic acid (RNA) is a polymeric molecule that is essential for most biological functions, either by performing the function itself (non-coding RNA) or by forming a template for the production of proteins (messenger RNA) [Wik25]. The goal of this project is to apply machine learning (ML) techniques to find and predict ncRNA-ncRNA interactions. Understanding these interactions is fundamental for an analysis of the **post-transcriptional regulation** and architecture of **cellular regulatory networks**. *Post-transcriptional regulation* is a set of processes that control gene transcription into proteins, while *cellular regulatory networks* are a representation of the complex web of interactions that control a cell's life. For medicine, building a strong understanding of how these processes work is fundamental to find the causes (and cure) of many diseases.

The challenge. Traditional computational methods, such as IntaRNA, estimate interaction energies using simplifying thermodynamic assumptions, like Minimum Free Energy (MFE) and accessibility models. While effective in specific contexts, these models are built on predefined parameters. This reliance on rigid, handcrafted rules fundamentally constrains their generalization capacity.

The solution. The poor generalization of traditional methods has driven a shift toward data-driven deep learning frameworks. These newer models can learn complex patterns directly from raw nucleotide data, avoiding the simplifying assumptions that limit older methods.

The goal of this project. The goal of this project is to implement a deep learning model (RNA-FM + FFNN) to predict the interactions between specific classes of ncRNAs. In particular, the interaction assigned to investigate is circRNA-miRNA.

The assigned interaction. This project's dataset focuses on the interactions between circular RNAs (circRNAs) and microRNAs (miRNAs). Unlike linear RNA, circRNAs are single-stranded RNAs that form a closed, continuous loop by joining their 3' and 5' ends. circRNAs can be non-coding regulatory regions or protein-coding regions. miRNAs are small, single-stranded, non-coding RNA regions involved in RNA silencing and post-transcriptional regulation of gene expression.

State of the art. The deep learning paradigm has found its maximum expression in the adaptation of architectures from Natural Language Processing (NLP), in particular Transformers [VSP+23]. The fundamental analogy is that the biological sequences (like proteins or RNA) can be treated as "language", where single nucleotides or amino acids act like "words" or "token" [NSJF+25, CHS+22]. This led to the development of powerful pre-trained models, known as RNA Language Models (RNA-LM). A prominent example, and central to this project, is RNA-FM, a Transformer-based foundational model trained on the large unannotated sequence database RNACentral [CHS+22]. As demonstrated by its authors, RNA-FM learns high-quality numerical representations (embeddings) that capture complex biological information, such as RNA structure and function, directly from the raw sequence [CHS+22] The actual state of the art consists in using these powerful pre-trained embeddings as input features for simpler classification models, like Feed-Forward Nerual Networks (FFNN). This approach uses the deep biologic knowledge "learned" by the language model to resolve specific tasks, like RNA interaction prediction.

2 Method

2.1 Dataset description (step 2)

The circRNA-miRNA interaction data is derived from RNA-KG, an ontology-based knowledge graph for representing interactions involving RNA molecules, built gathering more than 60 public databases [CCSG+24]. The raw dataset initially comprised **381** interaction pairs involving **379** unique RNA sequences. Figure 1 highlights the distinct size difference between these RNA types in the non-filtered dataset, illustrating that circRNAs are substantially longer, with a mean length of **1780.6** bp, compared to the much shorter miRNAs, which have a mean length of only **63.8** bp.

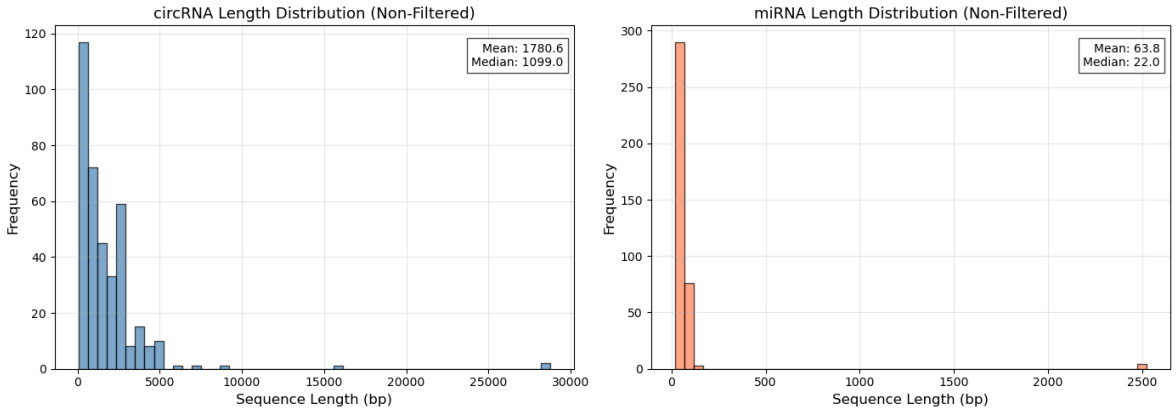


Figure 1: Sequence length distributions for circRNAs (left) and miRNAs (right) in the original, non-filtered dataset.

Preprocessing

To ensure data quality and model compatibility with RNA-FM [CHS+22], a two-step preprocessing pipeline was applied:

1. **Missing Value Removal:** Rows containing missing or null values were discarded, reducing the interaction count slightly to **373**.
2. **Sequence Length Filtering:** A strict length constraint was applied to ensure compatibility with the pre-trained RNA-FM encoder, which has a maximum input capacity of 1022 tokens. This step resulted in a significant reduction of the dataset, as a large portion of circular RNAs exceeded this limit.

After these filtering steps, the final dataset available for training consisted of only **159** positive interaction pairs.

Data Augmentation

Data augmentation is a fundamental step, which with a small effort, can lead to a significant improvement in the model performance. In this case for each interaction couple a 4X augmentation has been applied:

1. the original interaction $(seqX_i, seqY_j)$
2. sequence order inverted $(seqY_j, seqX_i)$
3. flipped sequences $(seqX_i^F, seqY_j^F)$
4. inverted and flipped sequences $(seqY_j^F, seqX_i^F)$

These operations increase the number of positive samples in the dataset, giving the model more data to learn from, and teaches the model the concept of "symmetry", which mean that the model understand that the sequences and nucleotides order doesn't matter, improving the generalization.

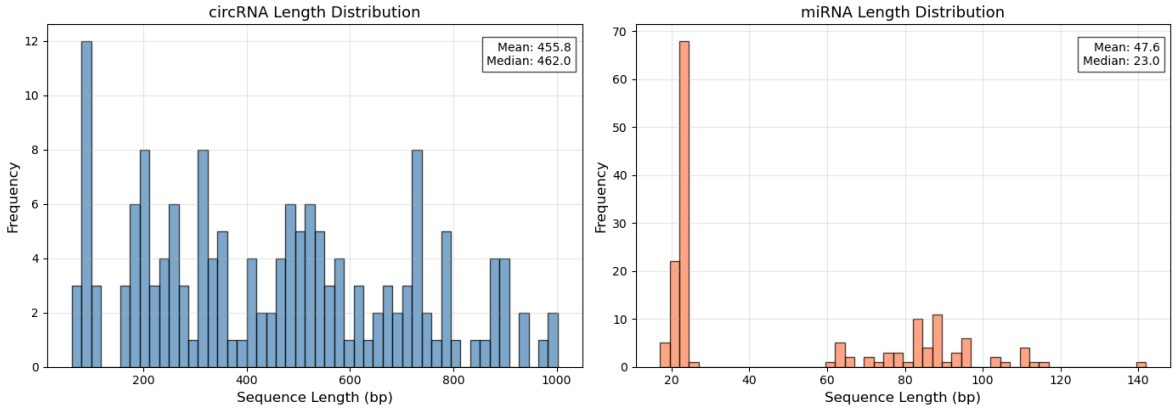


Figure 2: The two histograms show cirRNA and miRNA sequence length distribution after filtering

Negative Sampling

Negative sampling is another fundamental step as the dataset is completely unbalanced, containing only *positive* interactions. A binary classifier cannot be trained solely on positive examples; it must also learn to identify what constitutes a *negative* (non-interacting) case. To address this, we generate negative samples based on the known positive pairs. For each positive interaction (r_i, r_j), we create n negative pairs by keeping r_i and replacing r_j with a randomly selected sequence r'. The selection of r' is constrained by three rules:

1. $r' \neq r_i$ (Distinct Identity).
2. $\varphi(r') = \varphi(r_j)$ (Type Consistency, e.g., a miRNA is replaced only by another miRNA).
3. (r_i, r') is not a known positive pair (Novelty).

We set $n = 20$, intentionally creating a class imbalance (20:1 ratio) that forces the model to learn the highly specific features of a true interaction, mirroring the rarity of these events in reality.

Graph visualization (step 3)

The graph analysis reveals a sparse topological structure within the circRNA-miRNA interaction network. As illustrated in Figure 3, the network comprises 379 nodes connected by only 340 edges. The degree distribution is heavily right-skewed, characterized by a low mean degree of 1.79.

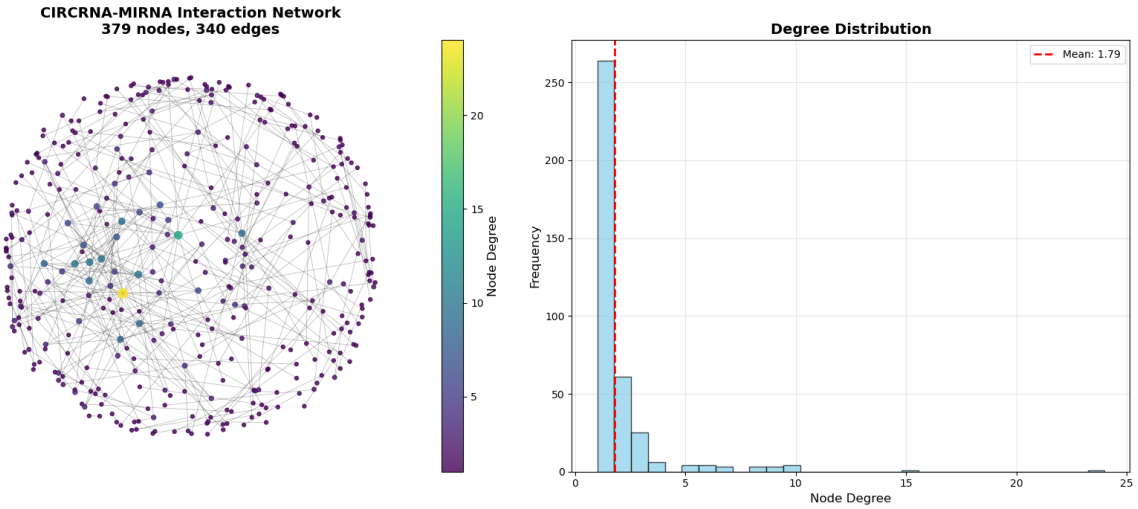


Figure 3: Graph analysis of the circRNA-miRNA dataset before filtering .

The application of the length constraint (< 1022 bp) significantly fragmented the network topology. As shown in Figure 4, the filtered network shrinks to 245 nodes and only 157 edges. The connectivity is further degraded compared to the original graph, with the mean degree falling to 1.28. This sharp reduction indicates that a substantial number of interactions in the original dataset involved large circular RNAs. The resulting structure is highly disconnected, consisting primarily of isolated pairs.

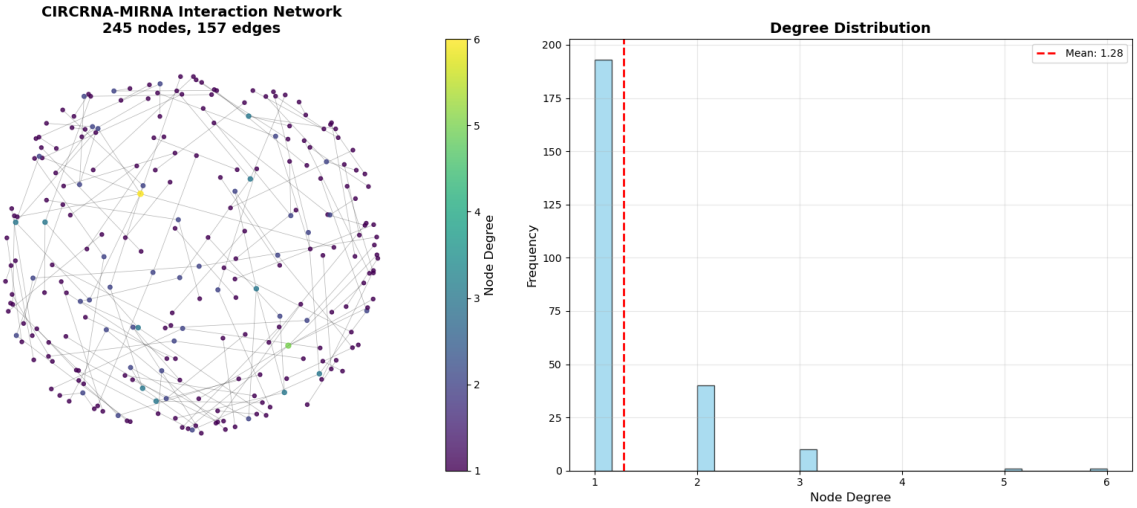


Figure 4: Graph analysis of the circRNA-miRNA dataset after filtering.

2.2 Model Architecture

The model architecture is well described by the schema in Figure 5: for every couple of ncRNA sequences the model generate their embeddings using the RNA-FM model, than these embeddings are aggregated and processed by a Feed-Forward Neural Network (FFNN) to compute the interaction probabilities.

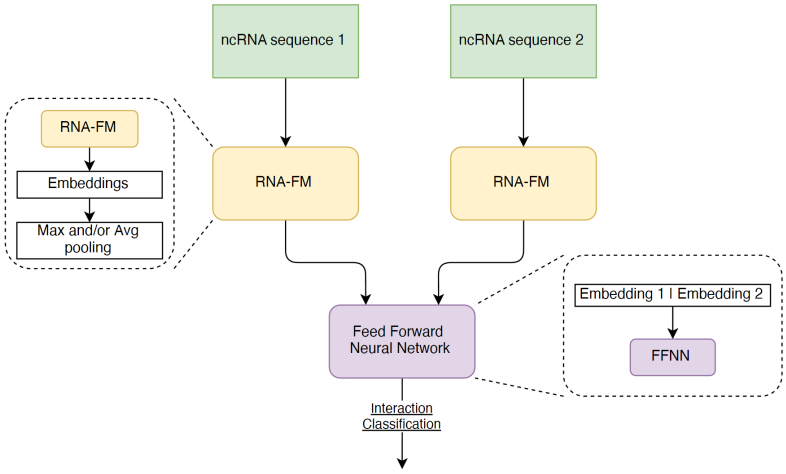


Figure 5: Model architecture schema

RNA-FM encoder

RNA-FM is a foundational language model built upon the BERT architecture, consisting of a deep stack of 12 Transformer encoder layers [CHS+22]. Designed to extract comprehensive structural and functional information without relying on experimental labels, the model undergoes self-supervised

pre-training on the RNAcentral database, which comprises over 23 million non-coding RNA (ncRNA) sequences [CHS+22].
The training utilizes a 'masked-token prediction' objective, where a percentage of nucleotides are artificially masked. By predicting their identity based solely on the bidirectional context of surrounding tokens, RNA-FM internalizes the underlying syntax, structural dependencies, and evolutionary patterns of RNA from vast amounts of unannotated data [CHS+22].

Feature extraction

At this point the RNA-FM model produced the embeddings which are L x 640 matrices, where L is the sequence length and 640 the number of features. The problem is that the final classifier, the Feed-Forward Neural Network, needs a fixed length vector as input. To achieve that a pooling layer was used, in particular two pooling techniques have been applied:

1. Max-pooling: extract the maximum value between all the nucleotides for each feature
2. Avg-pooling: calculate the average value of all the nucleotide for each feature

Then the results of each pooling technique are concatenated. The final embedding, for each ncRNA sequence, is a vector with 1280 features.

Feed-Forward Neural Network (FFNN) classifier

The final component of the architecture is a Feed-Forward Neural Network (FFNN) designed to perform the binary classification task.
The interaction prediction process proceeds as follows:

1. **Concatenation:** The feature vectors of the two RNA sequences, denoted as v_A and v_B (each of dimension $d = 1280$), are concatenated to form a single joint representation vector x of dimension $2d$.
2. **Dense Layers:** This vector x is passed through a series of fully connected layers. Each layer applies a linear transformation followed by a non-linear activation function (ReLU) to capture complex dependencies between the two sequences.
3. **Prediction:** The final layer consists of a single neuron with a Sigmoid activation function, which outputs a probability score $P(y = 1|x) \in [0, 1]$.

2.3 Experimental setup

Train split

The dataset is split into 90% training and 10% test sets, 20% of the training set is used for validation.

Training

The loss function used for training is the **Binary Cross-Entropy**:

$$\text{BCE} = -\frac{1}{N}\sum_{i=1}^{N}\left[y_i \log(p_i) + (1 - y_i)\log(1 - p_i)\right]$$

It quantifies the difference between the actual class labels y_i (0 or 1) and the predicted probabilities p_i output by the model. The lower the binary cross-entropy value, the better the model's predictions align with the true labels.

Then in order to contrast the unbalanced data, that could easily lead to a bias toward the negative interactions (they are 20 times the positive ones) a **mini-batch balancing** approach, with a 50% negative and 50% positive ratio, is applied. Positives are sampled with replacement while negatives are drawn without replacement to ensure diversity.

Evaluation metrics

Given the balanced nature of the test dataset (1:1 ratio), we evaluate the model using standard binary classification metrics derived from the Confusion Matrix elements: True Positives (TP), True Negatives (TN), False Positives (FP) and False Negatives (FN). The selected metrics are:

- **Accuracy**: The overall proportion of correct predictions. In a balanced setting, this serves as a reliable global indicator of model performance.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

- **Balanced Accuracy**: The arithmetic mean of sensitivity and specificity ($TN/TN + FP$). It evaluates how well the model predicts both classes individually. While mathematically equivalent to standard Accuracy in this specific 1:1 dataset, it is included to demonstrate robustness against class bias.

$$\text{Balanced Accuracy} = \frac{1}{2}\left(\frac{TP}{TP + FN} + \frac{TN}{TN + FP}\right)$$

- **Recall (Sensitivity)**: Quantifies the model's ability to correctly identify actual positive interactions (avoiding false negatives).

$$\text{Recall} = \frac{TP}{TP + FN}$$

- **Precision**: Reflects the reliability of positive predictions by measuring the proportion of predicted positive interactions that are actually true positives.

$$\text{Precision} = \frac{TP}{TP + FP}$$

- **F1-Score**: The harmonic mean of Precision and Recall, providing a single metric that balances the trade-off between false positives and false negatives.

$$\text{F1} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

- **AUROC (Area Under the Receiver Operating Characteristic Curve)**: Assesses the classifier's discrimination capability across all possible decision thresholds, summarizing the trade-off between the True Positive Rate and the False Positive Rate.

- **AUPRC (Area Under the Precision-Recall Curve)**: A complementary metric to AUROC. Even in balanced datasets, AUPRC provides granular insight into the model's performance specifically regarding the positive class (interactions), which remains the primary focus of biological discovery.

2.4 Hyperparameter tuning

Five sets of hyperparameters for the FFNN model have been tested to analyze the trade-off between model complexity and generalization capability:

1. **Original baseline**: The baseline production configuration serving as a reference point. It utilizes a moderate network depth (4 layers) and standard regularization (dropout 0.2), representing a balanced trade-off between computational efficiency and model complexity.

2. **Capacity boost**: Designed to capture highly complex non-linear dependencies between RNA sequences. This configuration increases both network depth (5 layers) and width (1536 hidden units), accompanied by a longer training schedule (70 epochs) to allow the expanded parameter space to converge.

3. **Regularized dropout**: Focuses on preventing overfitting through noise injection. It employs a significantly higher dropout rate (0.35) and a stronger negative reduction factor, forcing the model to learn robust features rather than memorizing specific training examples.

4. **Reduce Capacity**: A strategy aimed at aggressive simplification to counter severe overfitting on small datasets. It imposes a strict informational bottleneck by drastically reducing depth (2 layers) and width (256 units), combined with maximum regularization (dropout 0.5) and a cautious learning rate (1×10^{-4}) to ensure stable convergence.

5. **Balanced negative focus**: Prioritizes the minority class (positive interactions) to improve Recall. By increasing the negative reduction factor (3), this configuration alters the training distribution to place greater emphasis on positive samples, reducing the risk of "missed detections" (False Negatives).

Parameter	**Baseline**	**Capacity Boost**	**Reg. Dropout**	**Reduce Capacity**	**Bal. Neg. Focus**
			Model Architecture		
Input Dimension	2560	2560	2560	2560	2560
Hidden Dimension	1024	**1536**	1280	**256**	1152
Num. Layers	4	**5**	4	**2**	4
Dropout Rate	0.2	0.25	**0.35**	**0.5**	0.3
			Training Dynamics		
Batch Size	512	512	384	512	512
Learning Rate (LR)	5×10^{-4}	3×10^{-4}	4×10^{-4}	$\mathbf{1 \times 10^{-4}}$	5×10^{-4}
Min LR	5×10^{-5}	3×10^{-5}	4×10^{-5}	5×10^{-5}	4×10^{-5}
			Scheduling & Early Stopping		
Epochs	50	**70**	60	50	65
Warmup Epochs	4	6	5	4	5
Patience	10	14	12	10	10
			Data Handling		
Neg. Red. Factor	1	1	**2**	1	**3**
Val. Split Ratio	0.20	0.20	0.18	0.20	0.25

Table 1: Hyperparameter configurations evaluated during model optimization.

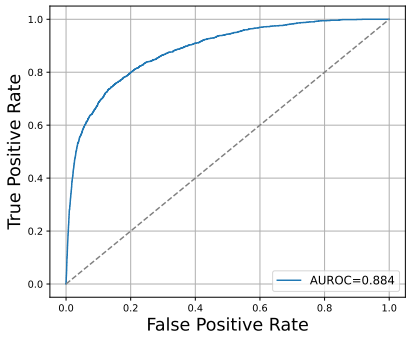
3 Results

Interaction	Accuracy	Bal. Acc.	Precision	Recall	F1-Score	AUROC	AUPRC
miRNA-miRNA	0.950	0.714	0.481	0.453	0.466	0.884	0.393
circRNA-miRNA	0.749	0.716	0.121	0.680	0.206	0.790	0.152

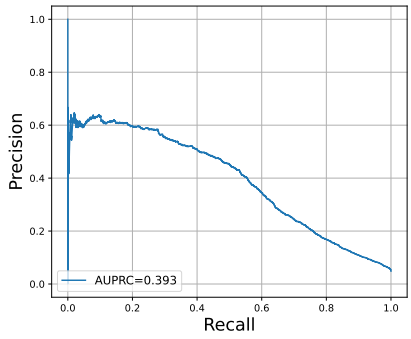
Table 2: miRNA-miRNA and circRNA-miRNA interactions metrics results with baseline parameters

3.1 Replication of miRNA-miRNA interaction classification task (step 1)

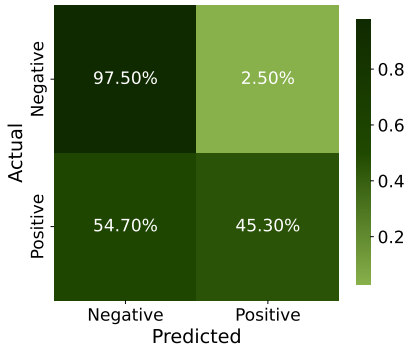
The training process exhibited healthy convergence as shown in Fig. 6d, with validation loss stabilizing around epoch 12, indicating effective learning without severe overfitting.
As shown in the performance plots, the model achieved a strong **AUROC of 0.884** (Fig. 6a), demonstrating good global discrimination capabilities. However, the **AUPRC of 0.393** (Fig. 6b) reflects the inherent difficulty of the highly imbalanced (1:20) dataset.
As shown in Table 2, detailed metric analysis reveals that while the global **Accuracy is 95.0%**, this is largely driven by the model's ability to correctly identify non-interacting pairs (True Negative Rate of 97.5%, see Fig. 6c). Prediction of positive interactions remains challenging, yielding a **Recall of 45.3%** and a **Precision of 48.1%**.
Overall, the RNA-FM embeddings provide a valid discriminative signal, confirming the architecture's readiness for the circRNA-miRNA target task.



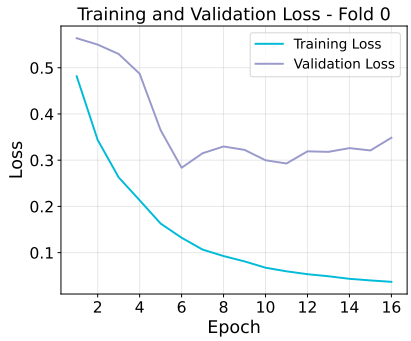
(a) ROC curve



(b) Precision-Recall Curve (PRC)



(c) Confusion matrix

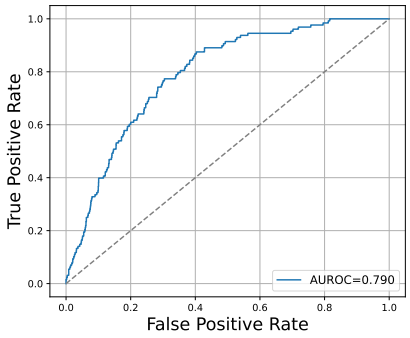


(d) Validation vs Training loss

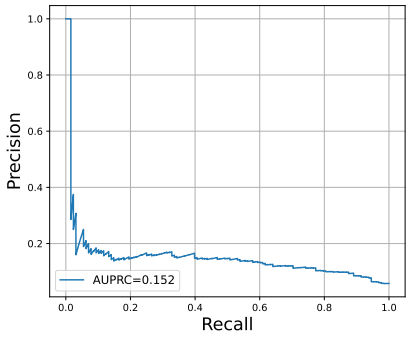
Figure 6: miRNA-miRNA interaction prediction model performance metrics.

3.2 Baseline hyperparameters for circRNA-miRNA classification (step 4)

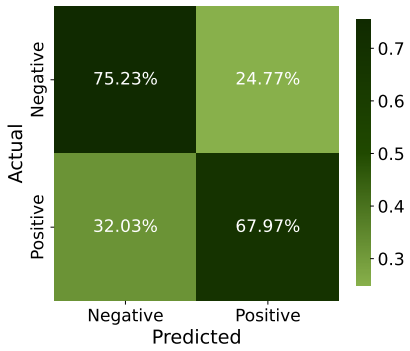
The training of the model on the circRNA-miRNA interaction dataset using baseline hyperparameters revealed significant generalization challenges. While the training loss demonstrated steady convergence, the validation loss diverged rapidly after the initial epochs, indicating severe overfitting (see Fig. 7d). As shown in the performance plots, the model achieved a respectable **AUROC of 0.802** (see Fig. 7a), suggesting it retains discriminative capacity to rank positive interactions higher than negative ones. However, the **AUPRC of 0.163** (see Fig. 7b) represents a critical bottleneck. Given the dataset's class imbalance (1:20 ratio), the random baseline precision is approximately 4.7%. Consequently, while the model performs roughly 3.5× better than random guessing (16.3% vs 4.7%), the precision is insufficient for a reliable screening tool, as it implies a high rate of false positives for every true detection.



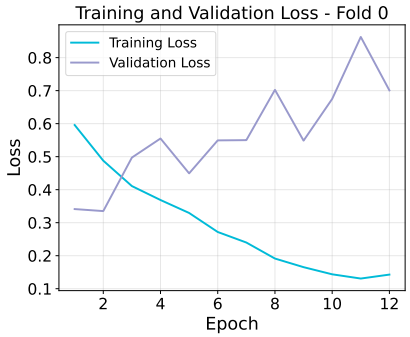
(a) ROC curve



(b) Precision-Recall Curve (PRC)



(c) Confusion matrix



(d) Validation vs Training loss

Figure 7: circRNA-miRNA interaction prediction model performance metrics, with baseline hyperparameters.

3.3 Hyperparameter tuning for circRNA-miRNA classification (step 5)

Experiment	Accuracy	Bal. Acc.	Precision	Recall	F1 Score	AUROC	AUPRC
Baseline	0.749	0.716	0.121	0.680	0.206	0.790	0.152
Capacity Boost	**0.788**	0.707	**0.133**	0.617	**0.218**	**0.801**	**0.167**
Reg Dropout	0.635	**0.742**	0.103	0.859	0.184	0.779	0.125
Reduce Capacity	0.518	0.710	0.084	**0.922**	0.155	0.794	0.144
Bal Neg Focus	0.652	0.717	0.100	0.789	0.178	0.781	0.152

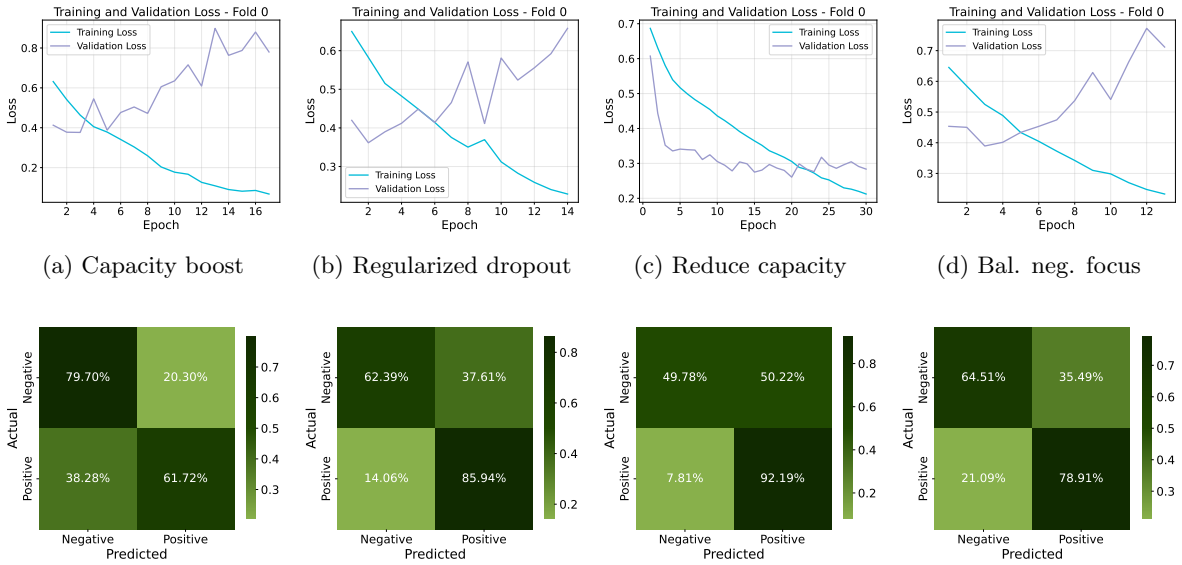
Table 3: Comparison of performance metrics across

The results of the hyperparameter optimization, summarized in Table 3, highlight the significant challenges in predicting circRNA-miRNA interactions. Among the configurations tested, the **Capacity Boost** model emerged as the most balanced approach, achieving the highest performance in key metrics, including precision (0.788), Precision (0.133), F1-Score (0.218), AUROC (0.801) and AUPRC (0.167). This suggests that despite the risk of overfitting, the complex interaction patterns encoded in the RNA-FM embeddings benefit from a model with higher representational capacity (deeper and wider layers).
In contrast, strategies aimed at aggressive regularization or simplification yielded mixed results. The **Reduce Capacity** and **Regulated Dropout** configurations achieved impressive Recall scores (0.922 and 0.859, respectively), successfully identifying the vast majority of true interactions. However, this came at a serious cost to Precision (< 0.11) and Accuracy. These models essentially defaulted to predicting "positive" for most uncertain pairs, leading to an unacceptably high False Positive rate.

Impact of Dataset Size. Crucially, the overall performance ceiling observed across all experiments—with AUPRC values consistently below 0.17 and F1-scores hovering around 0.2—points to a fundamental limitation beyond hyperparameter selection: **data scarcity**. Deep learning architectures, particularly those processing high-dimensional embeddings (2560 dimensions), typically require thousands of positive samples to establish robust decision boundaries. With a dataset limited to approximately 380 positive interactions, the models suffer from the "curse of dimensionality." The signal-to-noise ratio is too low for the classifier to learn precise distinguishing features, resulting in models that can generally rank positives higher (decent AUROC) but fail to make precise binary decisions (low Precision/AUPRC).

Visual Analysis of Training Dynamics. Figure 8 provides a clear view of how training progressed. The loss curves (top row) show the distinct impact of each strategy. The **Reduce Capacity** configuration (Fig. 8c) achieved the most stable training process. Its validation loss decreased steadily along with the training loss, confirming that simplifying the architecture effectively prevented the instability observed in other models. In contrast, the **Capacity Boost** (Fig. 8a) and **Balanced Negative Focus** (Fig. 8d) models show clear signs of overfitting, as their validation loss lines start rising sharply away from the training loss after epoch 10.

However, the confusion matrices (bottom row) reveal the cost of this stability. While the **Reduce Capacity** model had a stable loss, it developed a "high-recall bias" (Fig. 8g). It correctly identified 92.19% of positive interactions but failed to recognize over 50% of the negatives. This explains its very low Precision (0.084) in Table 3; the model essentially guessed "positive" for almost every uncertain pair. In contrast, the **Capacity Boost** model (Fig. 8e), despite its unstable loss curve, learned to distinguish classes better, maintaining a True Negative Rate of 79.70% while still capturing 61.72% of true interactions.



(a) Capacity boost (b) Regularized dropout (c) Reduce capacity (d) Bal. neg. focus

(e) Capacity boost (f) Regularized dropout (g) Reduce capacity (h) Bal. neg. focus

Figure 8: Hyperparameters tuning results for circRNA-miRNA classification task: the first row show the training and validation losses, the second row the confusion matrices.

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